

Run 10 | TEA

India

IV + controls

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 3,188 (11.57%) **Case time** 1m 04s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9975). Highest F1: CatBoost (0.9406). For XGBoost, adding controls changed ROC-AUC by +0.0003 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9975	0.9796	98.53	92.38	95.13	0.9374	<1s
CatBoost	0.9975	0.9783	98.60	92.42	95.76	0.9406	8s
H2O GBM	0.9975	0.9775	98.57	91.52	96.55	0.9396	3s
Random Forest	0.9973	0.9786	98.49	92.62	94.51	0.9355	1s
Logistic Regression	0.9973	0.9771	98.55	92.78	94.82	0.9379	2s
Ridge	0.9972	0.9751	98.58	92.03	96.08	0.9401	14s
Elastic Net	0.9971	0.9751	98.48	92.09	94.98	0.9351	14s
Lasso	0.9971	0.9747	98.42	91.79	94.82	0.9328	11s
Decision Tree	0.9940	0.9196	98.48	91.58	95.60	0.9355	<1s
Naive Bayes	0.9376	0.6793	92.72	64.50	82.42	0.7236	7s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	1,636.21	53.21
2	Established business owner	505.89	16.45
3	Occupational status	132.98	4.32
4	Age	67.05	2.18
5	Business angel	64.83	2.11
6	Knowing entrepreneurs	64.32	2.09
7	Business exit	54.73	1.78
8	Household size	42.90	1.40

Share is the percentage of total forest impurity importance across all 53 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Freedom to trade internationally	19.40	0.63
2	Overall economic freedom	17.09	0.56
3	Size of government	14.41	0.47

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 10_obj3_india_iv_cv_TEA. Scores and importance are taken directly from saved CSV outputs.